A uniformly bounded real normalization for the scalar elliptic detector

Abstract

We give a scalar detector whose real preparation is uniform in the elliptic point carrying the large index. The construction replaces an unbounded formal parameter by a bounded rational function and uses a positive rational normalization bounded away from zero. Its local valuation implications are unchanged outside an explicitly defined finite set of primes. This removes a real approximation requirement from the scalar certificate. It does not assert a reduction in the number of variables: prescribed indices must still satisfy the independent reduction-order conditions at the finite contact primes.

1 The elliptic curve and a bounded parameter

Put $K = \mathbb{Q}(\sqrt{2})$, $D = (3-\sqrt{2})^2$, and

$$E: \quad y^2 = x^3 - Dx.$$

Write σ for the nontrivial automorphism of $K/\mathbb{Q}$. An embedding of K into $\mathbb{R}$ will be denoted by ι. The identity component of $E_\iota(\mathbb{R})$ consists of the point at infinity and the affine points with $x \geq \sqrt{\iota(D)}$. In particular, every nonzero double has such an abscissa. For an affine non-torsion point define

$$f(Q) = \frac{1}{x(Q)}, \qquad z(Q) = -\frac{x(Q)}{y(Q)}, \qquad w(Q) = \frac{y(Q)}{x(Q)^2 + D}.$$

Lemma 1.1. *At either real embedding and every nonzero double Q,*

$$0 < \iota(f(Q)) \leq \frac{1}{\sqrt{\iota(D)}} < 1, \qquad |\iota(w(Q))| < 1.$$

The denominator $x(Q)^2 + D$ never vanishes at a K-point. Moreover,

$$\frac{w(Q)}{z(Q)} = -\frac{1 - Df(Q)^2}{1 + Df(Q)^2}. \tag{1}$$

At a good nonarchimedean place where Q lies in the formal group, the right side is a unit and is congruent to -1 in the residue field.

Proof. The assertion about f follows from the description of the real identity component and $3 - \sqrt{2} > 1$. For $x > \sqrt{D}$, evaluated at either embedding,

$$w^2 = \frac{x(x^2 - D)}{(x^2 + D)^2} < \frac{1}{x} \leq \frac{1}{\sqrt{D}} < 1.$$

The same bound holds at the endpoint, where $w = 0$. Total positivity of D proves nonvanishing of $x^2 + D$. Using $y^2 = x(x^2 - D)$ gives (1). At a good formal point, $v(f) > 0$ and D is integral, so the numerator and denominator on the right are units. □

Corollary 1.2. *Suppose P is a fixed non-torsion point and both $[c]P$ and $[h]P$ are in the formal group at a good place above a rational prime $p > 7$, with c, h nonzero integers. Then*

$$v\Big(\frac{w([c]P)}{w([h]P)}\Big) = v\Big(\frac{z([c]P)}{z([h]P)}\Big) = v(c/h).$$

Proof. The first equality follows from Lemma 1.1. For the second, the formal logarithm divided by z is a unit on the sufficiently deep formal neighborhood used in the scalar detector, and the formal logarithm is additive. At the large contact primes used here this is the usual first formal neighborhood. Equivalently, the two logarithms are nonzero integer multiples of the logarithm of a common nonzero formal multiple of P. □

2 A positive normalization bounded away from zero

Define the integer polynomials

$$\begin{aligned}\beta(X) &= 216X^4 + 450459,\\ \rho(X) &= \beta(X)(\beta(X)-1)(\beta(X)-2401)\\ &\qquad\cdot(32\beta(X)+343)(2\beta(X)-1).\end{aligned}$$

Put

$$\ell = 76249, \quad q = 1021, \quad a_0 = 8\cdot 108^5, \quad A_0 = 66\cdot 450459^5.$$

The letters ℓ and q denote two fixed primes. Both are avoidance primes for ρ: the accompanying exact certificate checks $\rho(x) \neq 0$ at every residue class modulo either prime.

2.1 A fixed polynomial denominator

Define

$$h_0(X) = 2239488X^{10} + 11663498286X^6 + \frac{9330368884708461}{512}X^2. \tag{2}$$

Direct expansion gives

$$\begin{aligned}F(X) := \rho(X) - 6h_0(X)^2 = &\frac{10882407499318210844052639}{64}X^8\\ &+\frac{1102703729539408567444895408037 81}{131072}X^4\\ &+1180712527837405404989927078052.\end{aligned} \tag{3}$$

Let N be the sum of these three positive coefficients. Let T be the smallest integer with

$$T \geq \max(\lceil N\rceil + 1, 2239488), \qquad T \equiv 2 \pmod{\ell}.$$

Explicitly,

$$T = 11815539940553962761175538801 04.$$

Set

$$\begin{aligned}&h = h_0 + T, \qquad H = 512h \in \mathbb{Z}[X], \qquad \delta = \rho - 6h^2 \in \mathbb{Z}[1/2][X],\\ &J = \mathrm{Res}_Y(\widetilde{\rho}, \widetilde{H}), \qquad D_0 = \left\lceil \sum_i |[X^i]\delta| \right\rceil, \qquad \mu = \frac{3\cdot 2239488}{8}.\end{aligned} \tag{4}$$

Here $\widetilde{\rho}, \widetilde{H} \in \mathbb{Z}[Y]$ are uniquely defined by $\rho(X) = \widetilde{\rho}(X^2)$ and $H(X) = \widetilde{H}(X^2)$. These quantities are fixed; none depends on the later exponent L.

Lemma 2.1. *The integer J is nonzero. For every real X,*

$$\delta(X) < 0, \qquad \mu(1+X^2)^5 < |\delta(X)| \le D_0(1+X^2)^5,$$

and

$$\frac{a_0}{D_0^2} < \frac{\rho(X)}{\delta(X)^2} < \frac{A_0}{\mu^2}. \tag{5}$$

The reduction of δ has no root modulo ℓ and its degree-ten leading coefficient is an ℓ-unit.

Proof. Put $\psi = 1 + X^2$. As $X^4 + 1 \ge \psi^2/2$,

$$108\psi^2 \le \beta(X) \le 450459\psi^2, \qquad a_0\psi^{10} < \rho(X) < A_0\psi^{10}.$$

All coefficients of h_0 and F are nonnegative, and $F(X) \le N\psi^4$. Hence

$$0 < \sqrt{\rho(X)/6} - h_0(X) = \frac{F(X)}{6(\sqrt{\rho(X)/6} + h_0(X))} \le \frac{N}{\sqrt{6a_0}\,\psi} < N.$$

It follows that $h - \sqrt{\rho/6} > T - N \ge 1$. Thus

$$|\delta| = 6(h - \sqrt{\rho/6})(h + \sqrt{\rho/6}) > 6h.$$

The inequalities $T \ge 2239488$ and $(1+X^2)^5 \le 16(1+X^{10})$ imply $6h \ge \mu\psi^5$. The coefficient norm of the degree-ten polynomial δ gives the upper bound for $|\delta|$. These facts prove (5).

If $\rho(\alpha) = 0$, then α^4 is a negative rational number, so α^2 is purely imaginary. Formula (2) shows that $h_0(\alpha)$ is purely imaginary. Since $T > 0$, $H(\alpha) \ne 0$. Therefore $J \ne 0$. Finally the two assertions modulo ℓ are checked on all 76249 residue classes in the supplied certificate, using $T \equiv 2 \pmod{\ell}$. The leading coefficient is $-12 \cdot 2239488\,T$, which is an ℓ-unit. □

Lemma 2.2. *For every integer X and every prime $p \ne 2, 3$, if $v_p(\rho(X))$ is odd, then*

$$v_p(\delta(X)) \le 2v_p(J).$$

Consequently $J^2\rho(X)/\delta(X)^2$ has no negative odd valuation at any prime $p \ne 2, 3$.

Proof. The polynomials $\widetilde{\rho}, \widetilde{H}$ are integral. Their resultant identity, evaluated at $Y = X^2$, implies

$$\min\{v_p(\rho(X)), v_p(H(X))\} \le v_p(J).$$

For $p \ne 2, 3$ the denominator 512 and the coefficient 6 are units. If $u = v_p(\rho(X))$ is odd and $v = v_p(h(X))$, the two terms in $\delta = \rho - 6h^2$ have different valuation parity. Thus $v_p(\delta) = \min(u, 2v)$. Put $j = v_p(J)$. If $u \le j$, the desired bound for $v_p(\delta)$ is immediate; otherwise $v \le j$, which gives the same bound. To obtain the sharper square multiplier, observe that

$$v_p(J^2\rho/\delta^2) = 2j + u - 2\min(u, 2v).$$

If $u \le j$, this is at least $2j - u > 0$. Otherwise $v \le j$. When $u < 2v$, oddness gives $u \le 2v - 1$, so the displayed valuation is at least $2(j-v)+1$. When $u > 2v$, it is again at least $2(j-v)+1$. The case $u = 2v$ is excluded by parity. If u is even, the valuation is even, proving the last assertion. □

2.2 The normalization and its local properties

Write

$$U_0 = \frac{A_0}{\mu^2}, \qquad c_* = 102^6 J^2, \qquad R_{\min} = \frac{a_0}{D_0^2 q^2 U_0}.$$

For each odd $L \geq 3$, let $k(L)$ be the smallest nonnegative integer satisfying

$$q^{2k(L)} > c_* \ell^L U_0,$$

and put

$$R_L(X) = \frac{c_* \ell^L \rho(X)}{q^{2k(L)} \delta(X)^2}. \tag{6}$$

Proposition 2.3. *For every real X and every odd $L \geq 3$,*

$$0 < R_{\min} < R_L(X) < 1.$$

The constant $R_{\min}$ is independent of L. For every rational X,

$$v_\ell(R_L(X)) = L + 2v_\ell(J). \tag{7}$$

For every integer X, the reduced denominator of $R_L(X)$ is a square, $R_L(X)$ is a square in each of $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and

$$v_2(R_L) \geq 4, \qquad v_3(R_L) \geq 4, \qquad v_{17}(R_L) \geq 4.$$

If $p \nmid 2 \cdot 3 \cdot 17 \cdot q\ell J$ is a positive contact prime of $\rho(X)$, then $v_p(R_L(X)) = v_p(\rho(X))$.

Proof. Since $U_0 > 1$ and $c_* \ell^L > 1$, one has $k(L) \geq 1$. Its minimality gives

$$\frac{1}{q^2 U_0} \leq \frac{c_* \ell^L}{q^{2k(L)}} < \frac{1}{U_0}.$$

Combine this with (5) for the real assertions. At an ℓ-integral rational X, both $\rho(X)$ and $\delta(X)$ are units. If $v_\ell(X) < 0$, their unit leading coefficients give $v_\ell(\rho(X)) = 20v_\ell(X)$ and $v_\ell(\delta(X)) = 10v_\ell(X)$. Both cases yield (7).

For integral X, direct residue calculations give

$$\rho(X) \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \quad v_2(\rho) = v_3(\rho) = 2, \quad v_{17}(\rho) \in \{0, 2\}.$$

At 2 and 3 the valuation of $6h(X)^2$ is odd, whereas the valuation of $\rho(X)$ is even. Thus $v_p(\delta) \leq v_p(\rho) = 2$. At 17, the number 6 is a nonsquare unit; since $\rho(X)$ is a nonzero local square, cancellation is again impossible in $\rho - 6h^2$, and $v_{17}(\delta) \leq v_{17}(\rho) \leq 2$. The factor 102^6 therefore supplies the stated local depths. The number ℓ is a local square at $2, 3, 17$, and all remaining factors in (6) are rational squares, so the local squareclasses are as claimed.

At $p \neq 2, 3, q, \ell$ apply Lemma 2.2; the additional factors are squares and do not lower the valuation. At q, $\rho(X)$ is a unit by the avoidance certificate, so $v_q(R_L)$ is even. At ℓ it is positive by (7). At $2, 3$ it is even and positive, as already proved. Thus no negative odd valuation occurs, proving the square-denominator assertion. Finally, at a positive contact prime outside the displayed fixed set, the resultant identity makes $H(X)$ a unit. The relation $\delta = \rho - 6h^2$ then makes $\delta(X)$ a unit, which proves the last assertion. □

3 Uniform real preparation of the norm tree

Let $\mathcal{U}$ be a fixed neighborhood in the compact real closure of the relevant multiples of P, chosen so that

$$|w(H)| \geq \eta, \qquad |\sigma w(H)| \geq \eta \quad (H \in \mathcal{U}) \tag{8}$$

for some fixed $0 < \eta \leq 1$. Such a neighborhood exists around any non-torsion multiple in the identity component of that closure. It is compatible with arbitrary fixed congruence conditions that select a finite-index subgroup: the closure of every such subgroup contains the identity component of the original compact closure.

Let $C_S = S - P$ be any nonzero double and $H \in \mathcal{U}$ a nonzero double. Write the rational components, in the basis $1, \sqrt{2}$, of

$$f(H), \qquad f(C_S), \qquad w(C_S)/w(H)$$

as $g_0, \ldots, g_5$. Set $m_0 = \cdots = m_3 = 1$ and $m_4 = m_5 = 20$. Fix an integer K such that

$$\ell \nmid K, \qquad K \geq \frac{4}{\eta \min(R_{\min}, R_{\min}^{20})}. \tag{9}$$

For $R = R_L(X)$ define

$$u_i = \frac{g_i}{K R^{m_i}}, \qquad N_R(u, v) = u^2 + Rv^2,$$

and the same norm tree as in the scalar detector:

$$w_0 = N_R(1, u_0), \quad w_1 = N_R(u_1, u_2), \quad w_2 = u_3, \quad w_3 = N_R(u_4, u_5),$$
$$W = N_R(N_R(w_0, w_1), N_R(w_2, w_3)).$$

The constants K and $R_{\min}$ are independent of L.

Theorem 3.1. *For every real X and every pair C_S, H satisfying the preceding conditions,*

$$1 \leq W \leq \frac{86249}{65536} < \frac{3}{2}, \qquad 0 < R_L(X) W^2 < \frac{9}{4} < 17.$$

These conclusions impose no real approximation condition on C_S. At every contact prime outside the fixed set of prime divisors of $2 \cdot 3 \cdot 7 \cdot 17 \cdot q\ell K J$, the local valuation implications of the scalar detector are unchanged by replacing z with w and dividing its six component tests by K. At the fixed prime ℓ, the detector gives the index depth

$$v_\ell(c) \geq 19L + 1, \qquad C_S = [c]P.$$

Proof. Lemma 1.1 and (8) bound the absolute value of each g_i by $1/\eta$: for a number $u + v\sqrt{2}$, its rational components are half the sum and $1/(2\sqrt{2})$ times the difference of its two real conjugates. The choice of K therefore gives $|u_i| \leq 1/4$. Since $0 < R < 1$,

$$1 \leq w_0 \leq 17/16, \quad |w_1| \leq 1/8, \quad |w_2| \leq 1/4, \quad |w_3| \leq 1/8,$$
$$1 \leq N_R(w_0, w_1) \leq 293/256, \qquad 0 \leq N_R(w_2, w_3) \leq 20/256.$$

Squaring these last bounds proves the asserted bound for W.

At a good consumed contact prime, K and $\delta(X)$ are units; the latter assertion follows from the resultant identity whenever the input is integral at that prime. Thus the contact valuation of R agrees with the prescribed contact valuation of ρ. At such a contact prime, the positive contact of $\beta(X)$ also implies that X is integral, since $p \nmid 216$. The order-one component tests put H, C_S in the formal groups exactly as before. Lemma 1.1 and

Corollary 1.2 then show that the two order-twenty tests have precisely the required ratio valuation.

More explicitly, if $\nu = v_\ell(R) > 0$ is odd, a node $u^2 + Rv^2$ has valuation equal to the smaller term valuation, because the two terms have different parity. The pole obstruction for RW^2 gives $v_\ell(W) > -\nu/2$. Following every path through the norm tree shows that each normalized leaf has valuation greater than $-\nu$. Since $\ell \nmid K$, both components of $w(C_S)/w(H)$ have valuation greater than 19ν. The exact formal ratio consequently gives $v_\ell(c/h) \geq 19\nu+1$. Since $h \in \mathbb{Z}$, Proposition 2.3 yields the claimed bound. □

Remark 3.2 (Finite exceptions and the contact-height constant). Suppose the contact-height theorem is written as $h(\beta) \leq HM(\beta)^{19}$, where $M(\beta)$ is the squarefree product of the consumed contact primes. Removing a fixed finite set F changes the bound only to

$$h(\beta) \leq H \left(\prod_{p \in F} p \right)^{19} M_F(\beta)^{19}.$$

Thus the finite exceptional primes of this normalization change a fixed constant and the resulting sufficient index threshold. They do not require an additional variable or an additional arithmetic existence theorem. No claim that the displayed constant or coefficients are small is intended.

3.1 Finite preparation is retained

The uniform real estimate removes only the real approximation condition on C_S. To apply the inherited pole-realization theorem to $R_L W^2$, the normalized leaves are still chosen sufficiently small at the fixed places above $2, 3, 17$. Their depths must absorb the valuation of K and the indicated powers of R_L. Choosing H and C_S sufficiently deep in these formal groups makes W a local unit, so that $R_L W^2$ retains the local squareclasses and depths of R_L. At every prime with positive odd $v_p(R_L)$ one likewise prepares the leaves so that W is a unit. At all remaining primes $v_p(R_L)$ is even, and multiplication by W^2 preserves evenness. Thus the reduced denominator of $R_L W^2$ is square. These are the same finite congruence choices supplied by the inherited scalar construction; they have not been replaced by the real bounds.

The height implication uses the five-contact theorem with contacts $\infty, 0, 1, 2401, -343/32$ and exponent 19, as stated in [1, Section 1] and its arithmetic dependency paper [3]. The input map makes the odd contact at infinity absent outside the fixed bad primes. Every other consumed contact is a positive contact of ρ, so Proposition 2.3 applies outside its fixed exceptional set. The additional factor $2\beta - 1$ does not add a hypothesis to that five-contact height theorem.

The numerator and denominator of R_L have degree twenty as rational functions of X. This observation concerns the scalar normalizer. It is not a new degree certificate for a compiled polynomial defining $\mathbb{Z}$.

4 The remaining finite-place condition

The real estimate does not replace the finite-place choice of the index c. If P has good reduction at a place $\mathfrak{p}$ and its reduction has order $d_\mathfrak{p}$, then

$$[c]P \text{ belongs to the formal group at } \mathfrak{p} \quad \Longleftrightarrow \quad d_\mathfrak{p} \mid c.$$

The integer $d_\mathfrak{p}$ can have prime factors different from the residue characteristic. Consequently $p^e \mid c$ does not by itself imply formal-group membership. In particular, a proposed prescribed

index

$$c = M\rho(X)^e$$

must still be shown to be divisible by the reduction order at every variable contact prime. The uniform real normalization proved here leaves this arithmetic requirement explicit.

Verification

The accompanying standard-library Python program

`check_uniform_scalar_normalization.py`

verifies the finite residue assertions, the exact norm-tree bound, and the constant inequalities. The resultant is evaluated independently by a Sylvester determinant and a product of five quadratic-factor formulas. The checker also verifies that the resultant in X equals the square of the smaller resultant in Y. The displayed proofs, rather than finite sampling, establish the assertions for all real or rational inputs.

References

[1] *A defining polynomial in twenty-seven variables: an existence theorem with a uniform degree bound*, project paper, version 12, 9 November 2026.

[2] *An explicit polynomial in twenty-nine variables with effective elliptic sum heights*, project paper, version 12, 9 November 2026.

[3] *Arithmetic dependencies of the construction with 29 variables*, project paper, version 8, 8 November 2026.

[4] *Scalar contact normalization by finite-field avoidance*, project paper, version 9, 8 November 2026.